Analysis of nucleotide position and orientation in 3D space reveals RNAs tendency for A and U unpaired nucleotides in close distance

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Introduction

RNA molecule:

- 3D structure determines function.
- Can bind to ligands, metabolites, etc.
- 3D structural dependencies tendencies need further exploration.

Goal

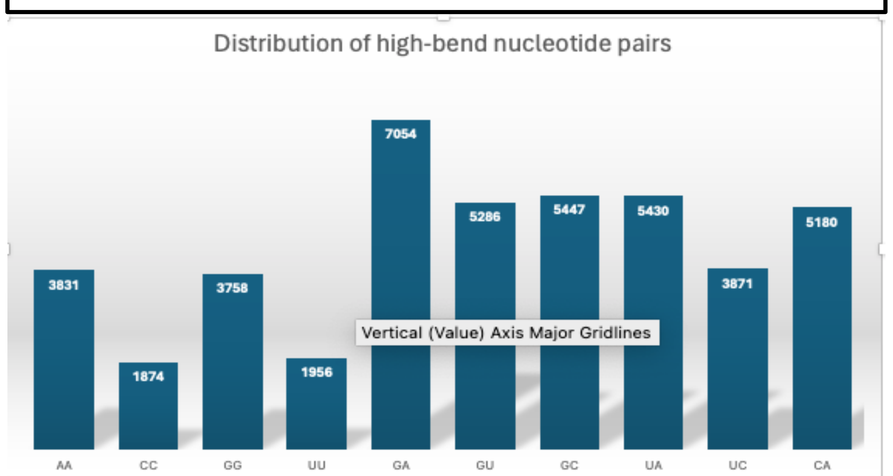
- Explore patterns of nucleotides that are in proximity in 3D space

Method

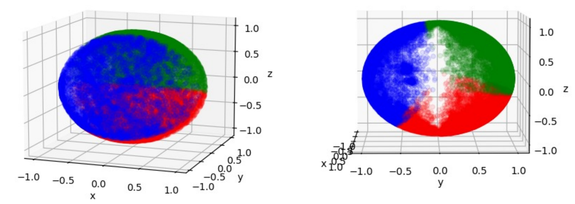
- Data
  - 13,000 PDB files, RNAsolo.
- Only unpaired nucleotides considered
  - Base pair elimination via x3dna-dssr
  - Clustering via DBSCAN
  - *Filtering criterion:*
    - Close in 3D (several Angstroms)
    - Far in sequence position

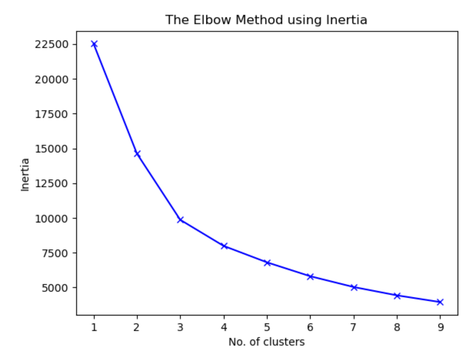
Results

- 43,000 high-bend pairs of nucleotides found
- (A and U), (A and A), and (G and A) were prevalent
- 43,687 high-bend nucleotide pairs found

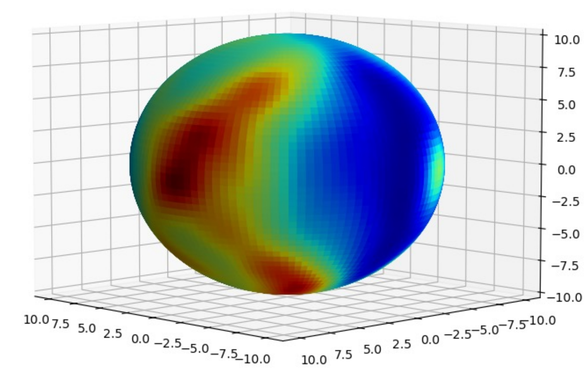


Distribution of relative orientation of high-bend unpaired nucleotides within RNA 3D structures





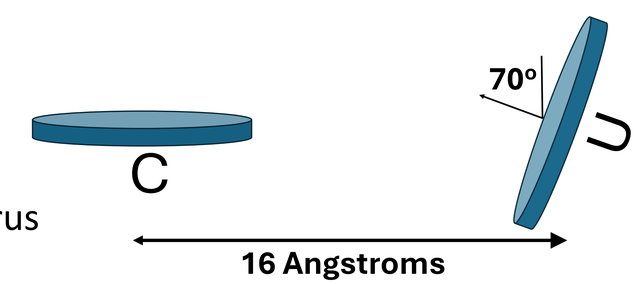
Density Plot

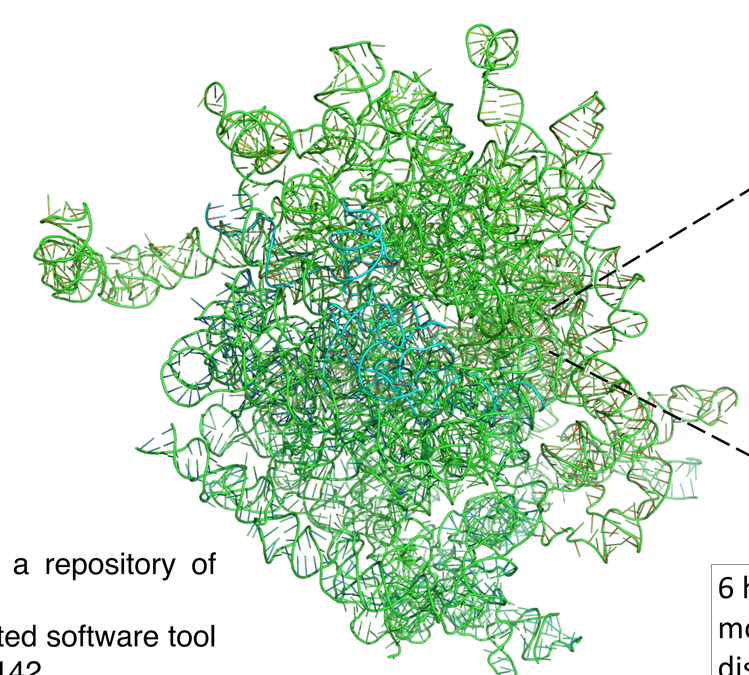


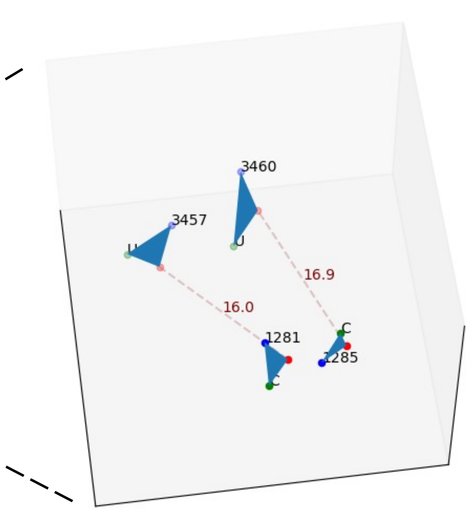
$$bend(i,j) = \frac{abs(i-j)}{dist3D(seq[i], seq[j])}$$

6ZVK|1|e2+h2

Oryctolagus cuniculus, Halastavi arva RNA virus







6 high-bend C and U pairs found in molecule:
distance = 16 Angstroms
relative orientation = 70°

References

- Bartosz Adamczyk, Maciej Antczak, and Marta Szachniuk. Rna-solo: a repository of cleaned pdb-derived rna 3d structures.Bioinformatics, 2022.
- Lu, X.-J., Bussemaker, H. J., and Olson, W. K. (2015) DSSR: An integrated software tool for dissecting the spatial structure of RNA. Nucleic Acids Res., 43(21), e142
- C. K. Joshi, A. R. Jamasb, R. Viñas, C. Harris, S. Mathis, and P. Liò. Multi-state rna design with geometric multi-graph neural networks. arXiv preprint, 2023b.